

The Committee Tax

Bullpen Entropy and the Cost of a Diverse Look

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Executive Summary

Every bullpen usage report tells a coach who pitched and when. None of them ask whether the bullpen, as a collective unit across a game, presents a coherent or chaotic look to the lineup facing it. This paper calls that collective look **Look Entropy**: how much a bullpen's release points, pitch shapes, and pitch-type mix vary across the arms a team deploys. Separately, it measures **Role Entropy**: how much a bullpen's pecking order shuffles from appearance to appearance, independent of what any arm looks like.

Across 310 qualified 2026 NCAA Division I bullpens, higher Look Entropy is associated with higher bullpen wOBA allowed ($R^2 = 0.101$, $p < 0.001$) and higher bullpen RA/9 ($R^2 = 0.084$). We call this the **Committee Tax**: a bullpen presenting a more varied collective look pays a real, moderate performance cost for that variety.

Role Entropy does not follow the same pattern. It is associated with outcomes in the opposite direction – pecking-order instability is associated with better results, not worse – and it is the largest wOBA change per one SD increase in the entire analysis. A second, independently-constructed measure, Pooled Tier Entropy, agrees with it. A bullpen can be visually diverse without being deployment-chaotic, and deployment-chaotic without being visually diverse. What a bullpen looks like and how it is deployed are two distinct forms of unpredictability, and they do not behave the same way.

This is one season of 2026 NCAA Division I baseball, and every finding here is observational: associated with, not shown to cause. The pitch-type classification model and empirical leverage index were trained on the full 2021-2026 NCAA Division I pool for stability; the shape-bin classification tree is trained only on seasons before 2026 and frozen, so 2026's own outcome never shapes 2026's own predictor (Methodology). Every result reported here describes 2026 specifically. A companion question – whether bullpen composition spills over into rotation performance – returned a result that does not clear conventional significance; that null (or near-null) finding is reported directly in Section 6, not reframed as something it isn't.

1 The Question

1.1 Why Bullpen Composition Might Matter

A hitter facing a bullpen across one game sees a sequence of different pitchers, not one continuous performance. Each new arm resets part of what the hitter has to read: release point, shape, pitch mix. The question this paper asks is simple to state and has not, to our knowledge, been asked formally at the college level: does the collective look a bullpen presents – taken as a whole, arm to arm across a game – predict how that bullpen actually performs?

1.2 From Pitcher-Level Anticipation to Bullpen-Level Entropy

This project is a direct structural sibling of the Anticipation Tax paper. That paper asked: what happens when a hitter’s expectation is disrupted from pitch to pitch, within one pitcher’s own arsenal? This paper asks a related but distinct question: what happens when a lineup’s expectation changes from pitcher to pitcher, across a bullpen, over the course of a game?

Anticipation Tax and this paper ask a similar question at different levels, but the two don’t scale the same way. Anticipation Tax found that pitch-to-pitch shock within one pitcher’s arsenal is associated with better outcomes for the pitcher creating it. This paper finds that arm-to-arm variety across a bullpen is associated with worse outcomes for the team presenting it, the opposite direction. Whatever mechanism makes within-pitcher unpredictability an asset does not obviously transfer to bullpen-level composition, and that distinction is one of this paper’s central findings.

1.3 What We Mean by a “Look”

A pitcher’s “look,” in this paper, is the combination of three things a hitter can perceive: where the ball comes from (release point), how it moves (shape), and what kind of pitch it is (type). A bullpen’s collective look is the variety of these three things across every arm the team actually uses. That variety, measured as entropy, is the core construct this paper builds on.

What a Bullpen "Looks Like" vs. How It's Deployed

Four pitchers create a collective look (Look Entropy) independent of who enters the game when (Role Entropy)

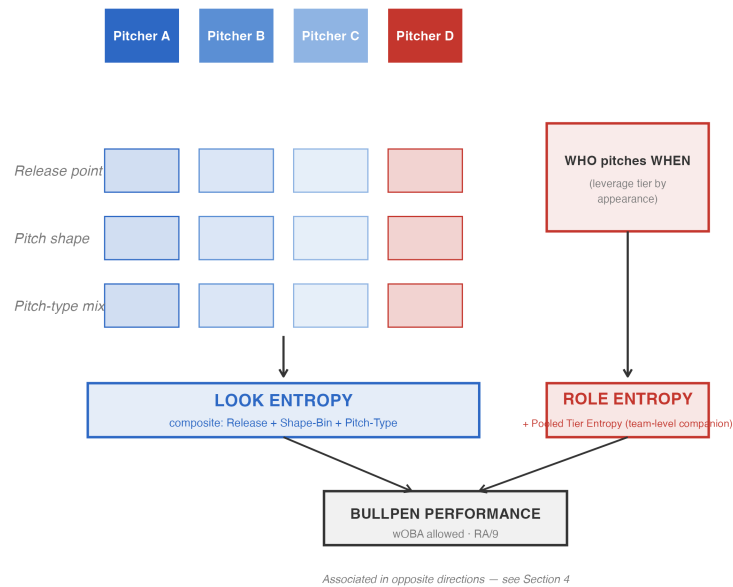


Figure 1: The entropy hierarchy: four pitchers each contribute a release point, a pitch shape, and a pitch-type mix. Their combined variety is Look Entropy. Deployment pattern – who pitches when – is a separate measure, Role Entropy.

The diagram above makes the distinction the rest of this paper depends on: the top half (what each pitcher looks like) feeds Look Entropy, and the bottom-right branch (who pitches when) feeds Role Entropy, entirely separately. The two are not different measurements of the same underlying thing – they describe different questions a hitter's brain has to answer, and Section 5 shows they behave differently in the data.

2 Measuring Bullpen Entropy

This paper measures bullpen entropy across four independent dimensions, plus one companion statistic. They are related but distinct, and should not be treated as interchangeable measures of “the same thing.”

- **Release Entropy:** how much release points vary across a bullpen’s arms, clustered into signature groups per handedness rather than treated as one continuous scale.
- **Shape-Bin Entropy:** entropy over data-driven pitch-shape bins, fit via regression tree per pitch type rather than hand-picked, whether a bullpen’s stuff looks similar or wildly different arm to arm.
- **Pitch-Type Entropy:** the simpler entropy of raw pitch-type mix across the bullpen.
- **Role Entropy:** not what an arm looks like, but when it gets used, entropy over which leverage tier a pitcher enters in, weighted by how often he actually appears. This captures pecking-order stability, a property of deployment, not of stuff.
- **Pooled Tier Entropy:** a companion to Role Entropy, not a fifth independent finding. Role Entropy asks whether an individual pitcher’s own tier-of-entry stays consistent across his appearances (per-pitcher, appearance-weighted). Pooled Tier Entropy asks how much tier variety the bullpen shows overall, pooling every appearance together regardless of which pitcher made it. A bullpen could show low Pooled Tier Entropy (mostly high-leverage innings) while individual pitchers’ own roles still shift within that.

The first three combine into a single composite, Look Entropy, with weights derived from each dimension’s standardized-predictor coefficient (wOBA change per one SD increase in that dimension) fit on seasons before 2026 and then frozen, so 2026’s own outcome never determines 2026’s own composite (Methodology, Data Quality Corrections item 7; the mechanism analysis below and the Sensitivity and Robustness appendix give the full derivation and its sensitivity to alternative choices). Role Entropy and Pooled Tier Entropy are reported separately throughout this paper, because Section 5 shows directly that they do not point the same direction as the Look Entropy trio.

3 Does Look Entropy Predict Bullpen Performance?

The most direct test of the Committee Tax idea is the simplest one: does a bullpen's Look Entropy score actually track its own performance? The figure below plots every qualified team's season-long Look Entropy against its bullpen wOBA allowed.

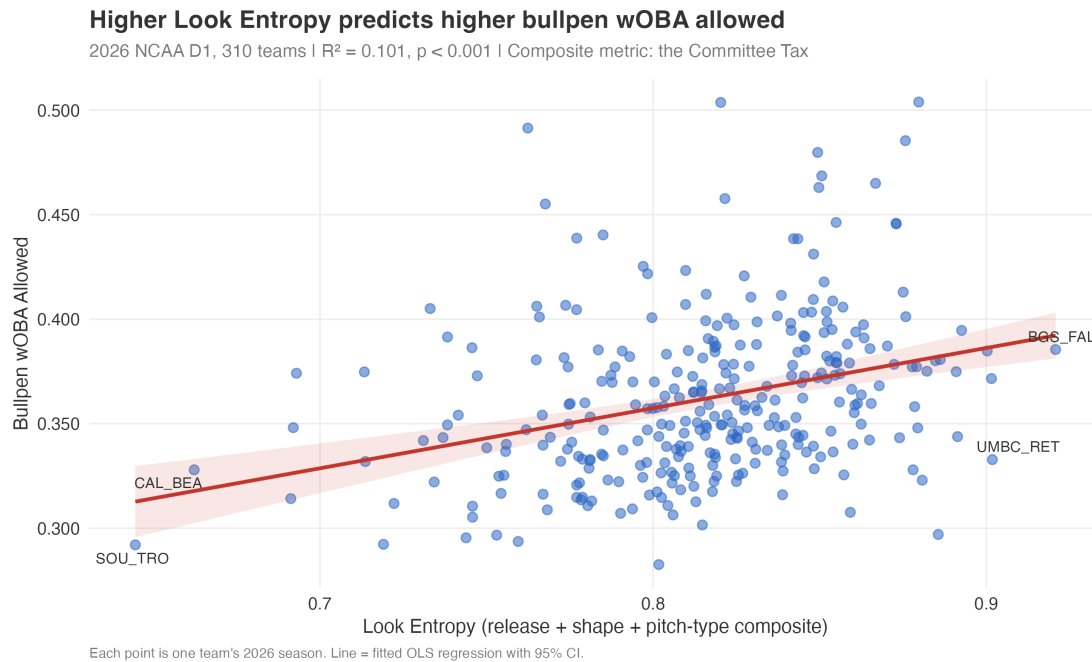


Figure 2: Look Entropy vs. bullpen wOBA allowed, 2026 NCAA Division I, qualified sample. Each point is one team's season; line is a fitted OLS regression with 95% confidence interval.

Across 310 qualified teams, Look Entropy is associated with bullpen wOBA allowed ($R^2 = 0.101$, $p < 0.001$) and bullpen RA/9 ($R^2 = 0.084$). Both relationships point the same direction: higher Look Entropy is associated with worse bullpen outcomes.

The headline result. A bullpen presenting a more varied collective look, across release point, pitch shape, and pitch-type mix, is associated with a worse bullpen wOBA allowed. The relationship is real and statistically clear, but it is one input among many: most of a bullpen's performance is explained by something other than how varied its arms look, which is what a reasonable prior would expect for a second-order effect layered on top of raw stuff quality.

3.1 Sample

Before trusting that number, here is exactly which teams it is based on.

Table 1: Sample inclusion criteria, 2026 NCAA Division I: 348 total team-seasons, 345 with a computable Look Entropy, 310 meeting the bullpen sample-size criteria.

Stage	N
Total 2026 NCAA D1 team-seasons	348
Excluded: no computable Look Entropy	0
Remaining: computable Look Entropy	348
Excluded: fewer than 5 relievers used	17
Excluded: fewer than 20 total bullpen innings (of those with enough relievers)	21
Final qualified sample	310

The full flow is 348 total 2026 D1 team-seasons, 348 of which have a computable Look Entropy at all, 310 of which also meet the bullpen sample-size floor used for every result in this paper. Fitting the same regression on the 348-team pool with a computable Look Entropy but no sample-size floor applied gives $R^2 = 0.006$, roughly 16.8x smaller than the qualified-sample result. These team-seasons are excluded because they are close to pure noise relative to entropy, not because the result is inconvenient: a handful of innings or relievers is not enough data for entropy to mean much, and noise mechanically drags a regression's R^2 toward zero regardless of how real the underlying relationship is in properly-covered data. The Sensitivity and Robustness appendix shows this result is stable across several reasonable alternative thresholds, not an artifact of the specific 5-reliever/20-inning cutoff chosen here.

Look Entropy is a composite of three separate dimensions, and they do not all behave the same way.

4 What Actually Drives the Relationship?

Testing each of Look Entropy's inputs independently, rather than only the composite, shows immediately that this is not one uniform effect. The figure below plots each dimension's own standardized coefficient against bullpen wOBA allowed, with its 95% confidence interval.

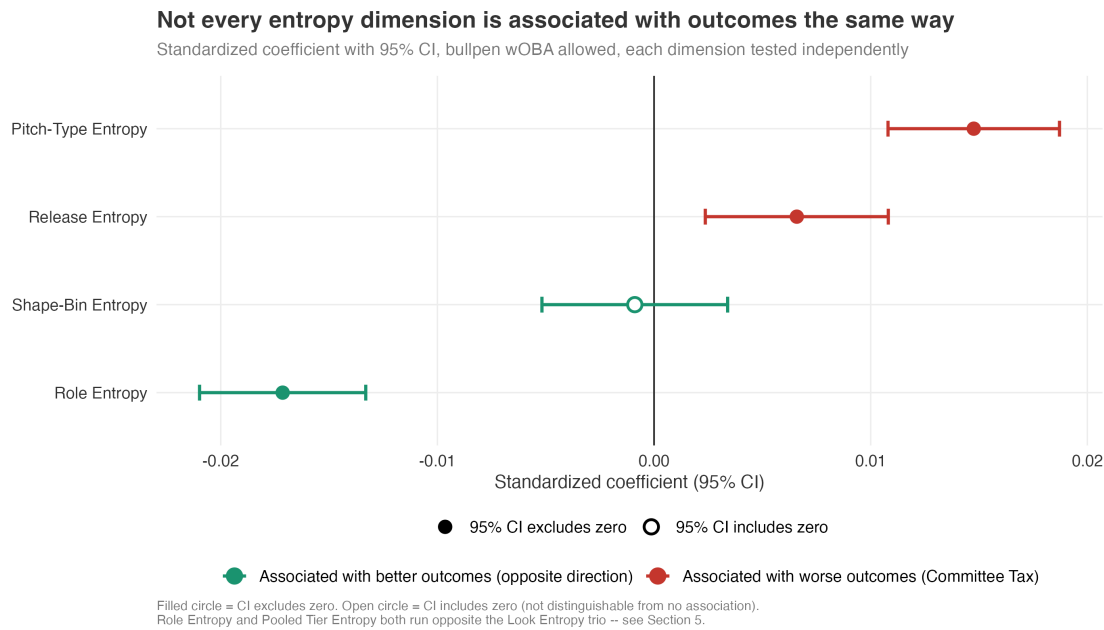


Figure 3: Standardized effect of each entropy dimension on bullpen wOBA allowed, with 95% confidence intervals, tested independently. Filled points: CI excludes zero. Open points: CI includes zero.

Two of the three Look Entropy inputs (Pitch-Type and Release Entropy) show a confidence interval that clearly excludes zero; the third (Shape-Bin Entropy) does not. Role Entropy and Pooled Tier Entropy, shown on the same chart for direct comparison, both run the opposite direction. The table below gives the exact numbers behind each point.

Table 2: Mechanism analysis. Each row is a separate regression predicting bullpen wOBA allowed from one standardized entropy dimension.

Mechanism	Estimate	95% CI	t	p	Direction
Release Entropy	+0.0066	[+0.0024, +0.0108]	3.06	0.002	Worse outcomes
Pitch-Type Entropy	+0.0148	[+0.0108, +0.0187]	7.31	< 0.001	Worse outcomes
Shape-Bin Entropy	-0.0009	[-0.0052, +0.0034]	-0.41	0.685	Better outcomes
Role Entropy	-0.0171	[-0.0210, -0.0133]	-8.76	< 0.001	Better outcomes
Pooled Tier Entropy	-0.0062	[-0.0105, -0.0020]	-2.89	0.004	Better outcomes

Note: Estimate is in raw wOBA units per one SD of the standardized predictor – bullpen wOBA has a small standard deviation (roughly 0.04), so even a real relationship produces a small-looking Estimate. R-squared (Figures 1-2) is the scale-invariant summary to compare across dimensions.

Pooled Tier Entropy is a companion statistic to Role Entropy, not part of the Look Entropy composite – see Section 2. These five dimensions and two outcomes (wOBA, RA/9) are tested independently with no formal multiple-comparison correction; most p-values here are well below any reasonable adjusted threshold, but this is worth stating explicitly given how much interpretive weight rests on significance for Shape-Bin Entropy specifically.

Three findings here deserve separate discussion rather than one paragraph glossing over all of them.

Pitch-Type Entropy shows the strongest, most direct effect of the three Look Entropy inputs. This is the most intuitive result: pitch type is an immediate, large perceptual difference for a hitter (fastball vs. slider vs. curveball changes trajectory, speed, and shape all at once), and a bullpen that varies this more from arm to arm is associated with meaningfully worse outcomes.

Release Entropy is significant but smaller. Release point is a real perceptual cue, but a subtler one than pitch type, and at a level where advance scouting is standard, hitters plausibly arrive with some prior exposure to a reliever’s release point before ever seeing a live pitch from him, which would partially dull whatever disruption effect exists. A smaller, still-significant coefficient is consistent with a real but secondary mechanism.

Shape-Bin Entropy is not statistically distinguishable from zero (95% CI includes zero, $p = 0.685$). This is worth stating plainly rather than minimizing: on this sample, whether a bullpen’s pitch shapes vary independent of pitch type and release point shows no detectable association with outcomes either direction. It still contributes to the Look Entropy composite (the next subsection addresses why, and the Sensitivity and Robustness appendix shows the composite is robust to removing it entirely).

4.1 The Look Entropy Composite

Given that one of the three inputs shows no detectable relationship on its own, the table below shows how much weight each dimension actually carries in the composite score used everywhere else in this paper.

Table 3: Look Entropy composite weights, derived from each dimension’s standardized-predictor coefficient fit on 2021-2025 (frozen; see Methodology, Data Quality Corrections item 7).

Dimension	wOBA per 1 SD	Weight
Release Entropy	0.0148	0.453
Pitch-Type Entropy	0.0125	0.382
Shape-Bin Entropy	0.0054	0.165

These weights are fit on seasons before 2026 and frozen, not recomputed from 2026's own data – so the coefficients in this table will not exactly match Table 2's, which is fit on 2026 specifically. That is intentional, not a discrepancy: Table 2 answers “does this dimension currently correlate with 2026 outcomes,” while this table answers “what weight was this dimension assigned before 2026 was ever seen.” Weights are derived from the magnitude of each dimension's standardized-predictor coefficient, not statistical significance, by design: significance is a function of sample size as well as coefficient magnitude, and gating weights on a p-value threshold would let an arbitrary sample-size difference between dimensions distort the composite. The tradeoff is visible directly in this table: Shape-Bin Entropy's small effect still receives a small (roughly 3% weight) rather than zero. The Sensitivity and Robustness appendix shows that excluding Shape-Bin Entropy entirely actually strengthens the composite's relationship with outcomes (R-squared moves from 0.101 to 0.1227, a 21% relative change): removing it does not weaken the case for the composite.

So far, every result has pointed the same direction: more variety in what a bullpen looks like is associated with worse outcomes. The next section breaks that pattern entirely.

5 Role Entropy Is Different

The chart below puts Look Entropy and Role Entropy side by side against the same outcome, so the contrast in Section 4's mechanism analysis is visible directly rather than requiring a reader to hold two separate figures in mind.

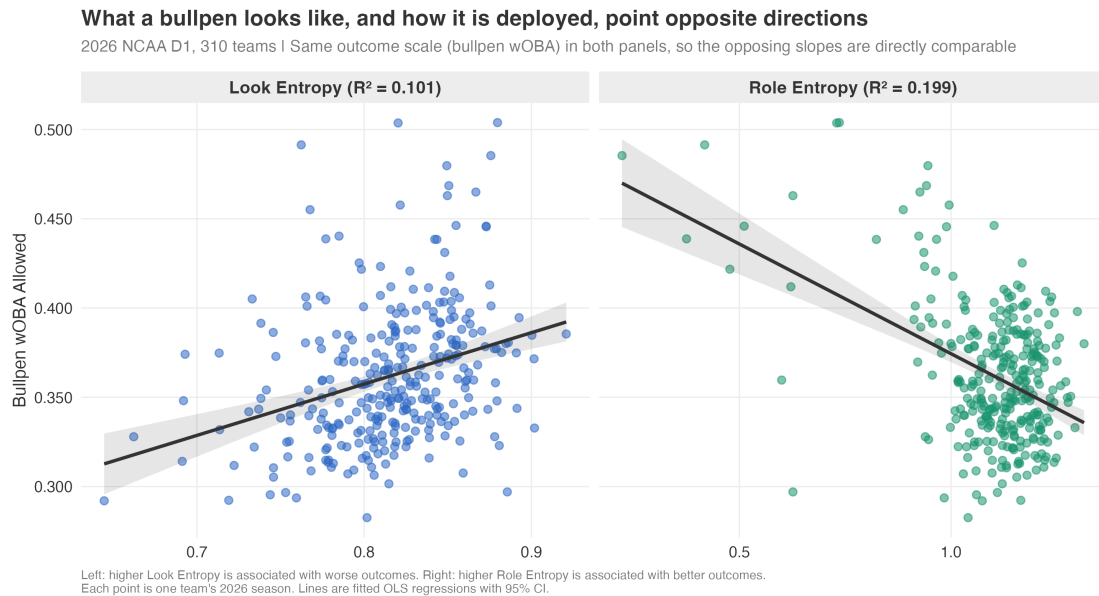


Figure 4: Look Entropy (left) and Role Entropy (right) vs. bullpen wOBA allowed, same y-axis scale in both panels. The two dimensions are associated with outcomes in opposite directions.

Role Entropy's coefficient is negative (-0.0171 , 95% CI $[-0.0210, -0.0133]$) and is the largest wOBA change per one SD increase in this entire analysis, larger than Pitch-Type Entropy's. Pecking-order instability is associated with better bullpen outcomes, not worse. Pooled Tier Entropy, a separate, independently-constructed measure of the same general idea (deployment variety, not look variety), agrees in direction. Two different ways of measuring deployment unpredictability both point the same way, and both point opposite the Look Entropy trio. That convergence is what makes this finding more than one measure's idiosyncrasy.

Bullpen unpredictability is not one thing. What a bullpen's arms look like (Look Entropy) and who gets deployed when (Role Entropy) are associated with outcomes in *opposite* directions. A team optimizing one should not assume it is learning anything about the other.

A bullpen can be visually diverse without being deployment-chaotic: a team could run four pitchers with wildly different release points and shapes, but always in the same fixed closer/setup/long-man order. A bullpen can also be deployment-chaotic without being visually diverse: four pitchers who all look similar, deployed in a constantly-shuffling order based on matchup rather than a fixed hierarchy. This analysis finds these two forms of unpredictability behave differently, and a coach optimizing for one should not assume it says anything about the other.

We do not claim to know the mechanism. One plausible reading: a bullpen that keeps hitters guessing about what is coming next pays a real cost for that unpredictability, while flexibility in who comes in when does not carry the same cost, and may reflect a coaching staff's ability to matchup-optimize rather than a hitter-facing disruption effect at all. This is speculative and stated as such; the data show the association, not the mechanism behind it.

6 Rotation Differential

Sections 3-5 all describe a bullpen's own outcomes. This section asks a separate, exploratory question: is bullpen composition associated with how a team's rotation performs, independent of the bullpen's own results? Team quality is the obvious confound here: programs with deeper pitching development plausibly produce both a more varied bullpen and a better rotation, which would show up as a relationship here without either one causing the other. The table below reports the relationship with and without a control for team pitching depth.

Table 4: Rotation wOBA against, regressed on Look Entropy, with and without a team pitching-depth control.

Model	Look Entropy coefficient	R-squared	p-value
Unadjusted	0.8498	0.064	0.044
Depth-adjusted	0.8282	0.064	0.116

This coefficient is not on the same scale as Table 2's mechanism estimates and should not be compared to them directly: Table 2 standardizes each predictor (one unit there is one standard deviation of that dimension), while the coefficient above regresses on raw, unstandardized Look Entropy, whose actual observed range across teams spans roughly 0.6 to 0.9. A one-point change in raw Look Entropy is far larger than any real team-to-team difference in this sample, so this coefficient describes a much smaller practical effect than its raw magnitude suggests.

The unadjusted coefficient reaches conventional significance ($p = 0.044$), but the depth-adjusted one does not ($p = 0.116$). We read this as informative rather than as a contradiction: a meaningful part of the unadjusted relationship appears to reflect program-quality confounding (teams with deeper pitching staffs tend to have both more varied bullpens and better rotations) rather than a standalone bullpen-to-rotation effect. Once that confound is controlled for, the evidence for a real spillover effect weakens substantially.

7 What Surprised Us

This section is deliberately about what did not match the working hypothesis going into this analysis, not a restatement of Section 4-5's numbers.

The direction of the headline finding itself. A bullpen presenting a wide variety of shapes and release points could plausibly make a lineup's job harder, the same logic Anticipation Tax found within a single pitcher's arsenal. Instead, for a bullpen as a collective unit, variety is associated with worse outcomes for the team presenting it. The pitcher-level finding does not scale up unchanged to the bullpen level, and this result is a reminder of that rather than a confirmation of it.

Role Entropy's reversal, and Pooled Tier Entropy's agreement with it. Going in, the working assumption was that any kind of bullpen unpredictability, what arms look like, or who gets deployed when, would create the same kind of disruption, in the same direction. Instead, deployment unpredictability and look unpredictability are associated with outcomes in opposite directions, and two independently-built measures of deployment unpredictability agree with each other.

8 Baseball Applications

Every application below is tied directly to a specific finding above; none of these are prescriptions beyond what the data actually show.

- **Bullpen construction.** If a program is choosing between two relievers of similar underlying quality but different pitch-type profiles, this analysis suggests weighing the team's existing pitch-type variety, not just the individual arm's quality in isolation; adding to an already-varied mix is associated with a modest additional cost per Section 4.
- **Scouting and advance prep.** Since Pitch-Type Entropy is the dimension with the clearest, largest signal, an opposing advance scouting report gains more from cataloguing a bullpen's pitch-type mix across arms than from release-point mapping alone, where the effect (while real) is smaller.
- **Deployment strategy.** Section 5's finding is an association, not evidence that matchup-flexible deployment causes better outcomes: a rigid, fixed bullpen hierarchy was not associated with better results in this sample, and a more matchup-flexible pattern was, if anything, associated with somewhat better ones.
- **Roster construction.** Shape-Bin Entropy's null result (Section 4) suggests that pitch shape diversity specifically, independent of type and release point, is not something to optimize for on its own; resources are better allocated toward pitch-type variety management.
- **What this does not support.** This analysis does not show that any specific reliever acquisition or lineup change causes a performance change; see Limitations. These are associations at the team-season level, not individual player evaluations.

9 Limitations

- **Single reporting season.** All headline numbers describe 2026 NCAA Division I specifically. The pitch-type classification model and leverage index were trained on 2021-2026 for stability; the shape-bin tree and Look Entropy weights are trained/derived only on seasons before 2026 and frozen (Methodology). The season-level entropy-outcome relationship itself has not been tested across multiple report years.
- **NCAA Division I scope.** This analysis excludes NCAA D2, D3, NAIA, JUCO, and any other level explicitly, enforced as a row-level filter described in the Methodology section below. Findings should not be assumed to generalize to other levels without separate testing.
- **Observational data; no causal identification.** Every relationship reported here is an association, not a demonstrated cause. Team quality, roster construction pressures, and unmeasured coaching decisions are all plausible confounds not fully separated out by these regressions.
- **Team-quality confounding.** A program that can afford to carry a varied bullpen may differ systematically in overall talent from a program that cannot, independent of whether variety itself does anything. This concern applies most directly to the Rotation Differential analysis (Section 6), and to a lesser but real degree to the main Look Entropy finding as well.
- **Missing baserunner state.** NCAA TrackMan data used here does not include baserunner state, so the empirical leverage index underlying Role Entropy is built from inning, top/bottom, outs, and score differential alone, a real but incomplete approximation of true win-probability leverage.
- **Imperfect leverage measurement and role classification.** Starter/reliever/opener classification and leverage-tier assignment both rest on empirically-derived thresholds (bootstrap-tested, not assumed) rather than ground-truth labels, and could misclassify some genuinely ambiguous appearances.
- **Sample-size sensitivity.** Section 3 and Appendix F show the core finding is stable across a range of reasonable inclusion thresholds, but any team-season-level analysis at this scale carries real sampling noise team to team.
- **Entropy depends on how states are defined.** Shape bins, release-point clusters, and leverage tiers are each the product of a specific construction choice (Section 10). A different, equally defensible construction could shift specific numbers, though Appendix F shows the qualitative pattern is not highly sensitive to the exact thresholds tested.
- **Rotation Differential is exploratory and underpowered** to detect a modest effect, if one exists; the null-to-marginal result in Section 6 should be read as “not detected here,” not “definitively absent.”

10 Methodology

10.1 Data and Division I Scope

NCAA TrackMan pitch-level data is restricted to NCAA Division I via an explicit configuration variable (`analysis_level <- "D1"`), applied as a row-level filter against each file's own recorded level, verified against the data itself rather than assumed from folder structure, before any cleaning, feature engineering, classification, or aggregation step touches the data. This is deliberately extensible: changing `analysis_level` to another value would restrict the same pipeline to a different level without rewriting the filtering logic itself. The two-tier scope from earlier work is preserved with one refinement: the pitch-type classification model and empirical leverage index are trained on the full 2021-2026 D1 pool, while the shape-bin classification tree and Look Entropy composite weights are trained/derived only on seasons before 2026 and then frozen, specifically so 2026's own outcome data cannot inform 2026's own predictors (Data Quality Corrections, items 6-7). Every player- and team-facing number in this report describes 2026 D1 specifically.

10.2 Data Quality Corrections

Several corrections were applied during development, each found by inspecting actual output rather than assumed away in advance:

1. **Pitcher identity canonicalization.** Inconsistent name formatting across source files was collapsing single real pitchers into multiple apparent identities; fixed by canonicalizing to the most frequent name variant per pitcher ID.
2. **Two bullpens per game.** An early pipeline version derived team identity for bullpen statistics from a game's first recorded pitch, but a game has two bullpens, and the away team always bats first, so this silently attributed both teams' bullpen activity to the home team. Fixed by using each pitch's own, already-correct team attribution.
3. **Handedness-normalization in pitch-shape features.** Two derived features used in pitch-type classification (a horizontal-break interaction term and a spin-axis-based term) were z-scored against a population pooled across both throwing hands. Since right-handed pitchers outnumber left-handed pitchers in any real D1 pool, this effectively calibrated both features to a right-handed population, causing systematically different classification behavior for left-handed pitchers on the same physical pitch shape. Fixed by hand-normalizing both features before scoring.
4. **Release-point-based handedness verification.** Raw PitcherThrows was found to be unreliable for a subset of pitchers. An independent, release-point-based handedness model (k-means clustering on release height and release side, majority-vote labeled against the raw field) is fit before any hand-dependent feature is built, and each pitcher's PitcherThrows is corrected to the consensus of that model across all of his own appearances, since a pitcher's physical release point and throwing hand do not change appearance to appearance.
5. **A sample-size floor.** Team-seasons with fewer than 5 relievers used, fewer than 20 total bullpen innings, or a missing composite score are excluded before any statistic in this report is computed (Section 3).
6. **Shape-bin target leakage.** The shape-bin CART (see Shape-Bin Classification below) was originally fit on the full pooled training population, including 2026's own per-pitch wOBA – meaning 2026's outcome data was indirectly shaping the very predictor later evaluated

against that same outcome. Fixed by fitting the tree only on seasons before 2026 and freezing it; 2026's pitches are assigned bins from that frozen tree but never contribute to fitting it.

7. **Outcome-derived Look Entropy weights.** The same issue applied one level up: Look Entropy's composite weights were derived from 2026's own effect sizes, and that composite was then evaluated against 2026's own outcome – an in-sample, outcome-derived predictor. Fixed the same way: weights are derived only from seasons before 2026, frozen, and then applied to construct Look Entropy for every season including 2026. The weighting philosophy itself (magnitude of each dimension's standardized-predictor coefficient, normalized to sum to one) is unchanged – only the population used to fit it.

10.3 Shape-Bin Classification

Pitch shape bins are fit via regression tree (CART) per pitch type, predicting a per-pitch wOBA outcome from velocity, induced vertical break, horizontal break, spin axis, and extension. Bin edges are the tree's own outcome-optimal splits, not hand-picked round numbers. The tree is fit only on seasons strictly before the reporting year and then frozen: 2026's own pitches are assigned bins from that frozen tree but never contribute to fitting it, so 2026's outcome data cannot indirectly shape 2026's own shape-bin predictor.

10.4 Entropy Calculation

Each entropy dimension is a Shannon entropy calculation, $H = -\sum_i p_i \log(p_i)$, over the relevant category distribution (pitch type, shape bin, release-point cluster, or leverage tier) within a bullpen's pitches or appearances. Release, Shape-Bin, and Pitch-Type Entropy are each normalized (divided by $\log(k)$, where k is the number of categories observed) so bullpens using different numbers of distinct categories remain comparable on a 0-1 scale. Role Entropy does not share this ceiling: it is an appearance-weighted average across each pitcher's own tier-of-entry, not a single normalized distribution, and its observed range in this sample runs from 0.223 to 1.314 (mean 1.101). Values above 1.0 are expected under this construction, not a normalization error, and Role Entropy should be read as its own scale throughout this paper rather than compared directly to the 0-1 dimensions.

10.5 Rarefaction

Game-level entropy calculations use rarefied Shannon entropy (resampled to a common reference pitch count across games being compared) rather than raw entropy, since raw entropy is mechanically biased upward by pitch volume alone. Season-level entropy uses raw normalized entropy instead, since sample-size disparities matter far less at that grain.

10.6 Empirical Leverage Index

Leverage tiers are derived empirically from this project's own 2021-2026 D1 win-probability data, inning, top/bottom, outs, and score differential, with no baserunner state available in NCAA TrackMan, rather than assumed to match MLB's leverage curve. Candidate tier boundaries were tested via bootstrap confidence intervals before being finalized.

10.7 Regression Methodology and Uncertainty Estimation

All mechanism-level regressions (Section 4's mechanism analysis) are simple OLS regressions of a single standardized entropy dimension against a single outcome, fit independently per dimen-

sion rather than jointly in one multivariate model; reported effect sizes should be read as each dimension's own association, not a partial effect controlling for the others. Standard errors are the standard OLS closed-form estimate; 95% confidence intervals are estimate $\pm 1.96 \times \text{SE}$.

11 Appendix

11.1 A. Pitch Movement Reference — Relief Pitchers Only, by Throwing Hand

The reference below is the baseline scouting chart this entire study’s shape and release measures are built from: every relief pitcher’s actual pitch movement, split by throwing hand so a lefty and a righty are never pooled onto one visually misleading scale.

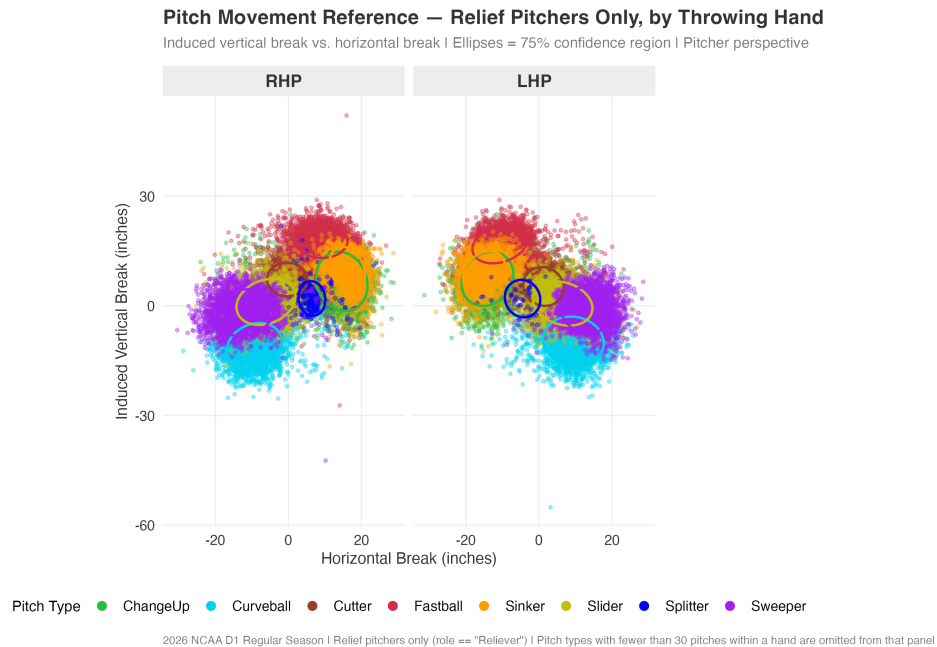


Figure 5: Pitch movement reference for relief pitchers only, split by pitcher throwing hand. Colors identify pitch type and are held constant across both panels. Ellipses = 75% confidence region per pitch type, computed separately within each hand; pitcher perspective.

Restricted to pitchers classified as relievers, since this entire study is about bullpen composition. Split by the pitcher’s own throwing hand, not batter handedness – a lefty and a righty with numerically similar raw horizontal break are not presenting the same look to a hitter, and pooling them would blur that distinction. Pitch types with fewer than 30 pitches within a given hand are omitted from that hand’s panel rather than forcing an unreliable confidence ellipse.

11.2 B-C. Average Pitch Shape by Hand

The movement reference above shows the shape of each pitch type’s distribution; the tables below give the actual numbers a reader would need to describe a specific pitch precisely rather than read approximate values off a scatter plot.

Table 5: Average pitch shape by pitch type, RHP relievers only.

Pitch Type	N	Avg Velo (mph)	Avg IVB (in)	Avg HB (in)
Fastball	493708	90.3	17.5	8.9
Slider	306482	81.7	1.1	-6.7
Sinker	209512	90.0	9.7	15.9
ChangeUp	171840	81.9	6.5	13.9
Curveball	96772	77.1	-11.0	-9.1
Cutter	78163	84.2	7.1	-0.2
Sweeper	41907	77.0	-1.7	-12.2
Splitter	271	83.7	2.2	6.9

Table 6: Average pitch shape by pitch type, LHP relievers only.

Pitch Type	N	Avg Velo (mph)	Avg IVB (in)	Avg HB (in)
Fastball	192885	88.9	16.6	-10.2
Slider	75610	80.1	0.7	6.5
ChangeUp	66918	80.4	7.5	-13.6
Sinker	52001	88.8	9.2	-14.9
Curveball	35017	75.8	-9.7	9.2
Cutter	32736	81.7	5.3	1.1
Sweeper	11659	75.8	-1.2	13.3
Splitter	48	82.7	1.9	-5.0

Read alongside Appendix A: a pitch type showing a large difference in average horizontal break between the RHP and LHP tables is expected and correct (that is the mirrored, hand-specific movement Section 10's handedness-normalization fix accounts for before any modeling step); a pitch type missing from one hand's table entirely means fewer than 30 qualifying relievers of that hand threw it in the 2026 season, not that the pitch type doesn't exist for that hand.

11.3 D. Additional Model Diagnostics

The Methodology section describes a handedness-reliability correction applied during data cleaning; the figure below is the actual diagnostic used to find and verify that fix, included here for full transparency rather than only summarized in prose.

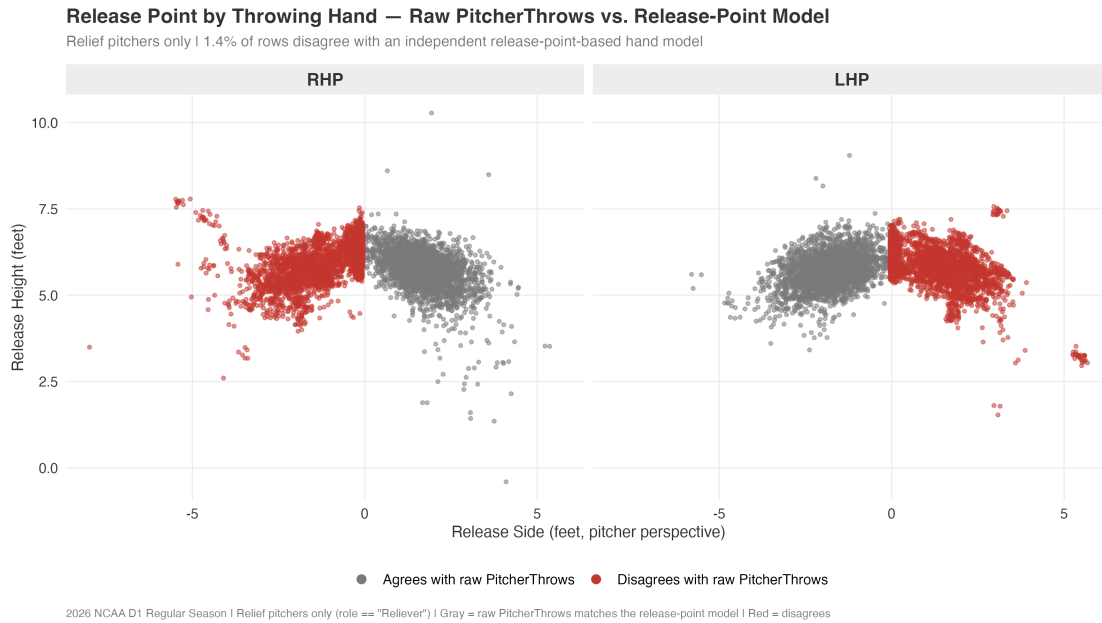


Figure 6: Release point by throwing hand, colored by whether an independent release-point-based hand model agrees with raw PitcherThrows. Used during development to verify handedness reliability; included here for transparency.

Gray points are rows where the raw recorded throwing hand agrees with an independently-fit, release-point-only hand model; red points disagree. A small number of disagreements is expected (real measurement noise in release point); a cluster of red points concentrated in one hand’s panel would indicate a systematic labeling problem rather than noise, which is exactly the pattern the Methodology section’s handedness correction was built to catch.

11.4 E. Additional Entropy Distributions

The main text reports each entropy dimension’s coefficient and composite weight; the table below gives the actual shape of each dimension’s distribution across qualified teams, for readers who want a sense of what a “typical” or “extreme” score looks like on each scale.

Table 7: Distribution of each entropy dimension, qualified 2026 D1 sample (N=310).

Dimension	Mean	5th pct.	95th pct.
Look Entropy	0.816	0.743	0.878
Role Entropy	1.101	0.922	1.242
Release Entropy	0.857	0.734	0.951
Shape-Bin Entropy	0.846	0.750	0.908
Pitch-Type Entropy	0.753	0.660	0.848

11.5 F. Sensitivity and Robustness Analyses

This section shows how sensitive the Section 3 finding is to reasonable analytical choices. Four checks are reported here.

Table 8: Threshold sensitivity: composite regression result across alternative minimum-reliever/minimum-inning cutoffs. The 5-reliever/20-inning row is what Section 3 reports.

Min. relievers	Min. innings	N	R-squared	p-value
0	0	348	0.006	0.153
3	10	314	0.096	< 0.001
5	20	310	0.101	< 0.001
8	30	305	0.122	< 0.001
10	40	304	0.123	< 0.001

Note: The 0/0 row applies no reliever or innings floor, but still excludes any team-season with no computable composite score at all – a small number of team-seasons fall into this category regardless of threshold, which is why this row’s N can be marginally below the raw team-season count reported in Section 3.

Table 9: Composite weighting sensitivity. Removing Shape-Bin Entropy strengthens the result (R-squared 0.101 to 0.1227); equal-weighting (ignoring effect size) weakens it.

Specification	R-squared	Corr. with current composite
Current (effect-size weighted)	0.1010	1.0000
Excluding Shape-Bin Entropy entirely	0.1227	0.9835
Equal-weighted (1/3 each)	0.0744	0.9693

Table 10: Extreme-team exclusion sensitivity: the composite relationship is not driven by a handful of outlier teams.

Specification	R-squared
Full qualified sample (N=310)	0.1010
Trimmed 5th-95th percentile by Look Entropy (N=278)	0.1075

Across every alternative specification tested, looser or stricter sample-size thresholds, excluding Shape-Bin Entropy from the composite, equal-weighting instead of effect-size weighting, and trimming the most extreme 10% of teams by Look Entropy, the qualitative finding (a positive, significant association between Look Entropy and bullpen wOBA allowed) is stable. Two specifications meaningfully move the R-squared: equal-weighting weakens the result, while excluding Shape-Bin Entropy entirely strengthens it – both are informative in the same direction, since they show effect-size weighting is doing real work relative to a naive equal-weighted alternative, and that Shape-Bin Entropy’s small weight is not propping up a result that depends on it.

The main text tests each of the five mechanism dimensions independently with no formal multiple-comparison correction. The table below applies a Benjamini-Hochberg FDR adjustment across those five tests to check whether that changes any conclusion.

Table 11: Benjamini-Hochberg FDR-adjusted p-values across the five independent mechanism tests in Table 2.

Mechanism	Raw p	FDR-adjusted p	Conclusion changes?
Release Entropy	0.002	0.004	No
Pitch-Type Entropy	< 0.001	< 0.001	No
Shape-Bin Entropy	0.685	0.685	No
Role Entropy	< 0.001	< 0.001	No
Pooled Tier Entropy	0.004	0.005	No

The FDR adjustment does not change any dimension’s significance conclusion: every result that was significant at $p < 0.05$ before correction remains significant after it, and vice versa. This is expected given how far most of these p-values sit from any reasonable threshold, but it is worth confirming directly rather than assuming it given how much interpretive weight rests on significance for Shape-Bin Entropy specifically.

11.6 G. Technical Definitions

- **Shannon Entropy:** $H = -\sum_i p_i \log(p_i)$, where p_i is the proportion of observations in category i .
- **Normalized Entropy:** $H/\log(k)$, scaling raw entropy to a 0-1 range given k observed categories.
- **Rarefaction:** resampling to a common reference sample size before computing entropy, correcting for the mechanical upward bias raw entropy has with sample size.
- **Look Entropy:** the composite of Release, Shape-Bin, and Pitch-Type Entropy, weighted by each dimension’s standardized-predictor coefficient fit on seasons before the reporting year and frozen.
- **Role Entropy:** appearance-weighted entropy over each pitcher’s own leverage-tier-of-entry across his appearances.
- **Pooled Tier Entropy:** entropy over leverage-tier-of-entry pooled across all of a bullpen’s appearances, regardless of pitcher.